Nishal Stanislaus Silva, *PhD*

Machine Learning Research Engineer | Speech, NLP & Multimodal ML | Applied Science

Canadian Permanent Resident, Eligible to work in Canada without sponsorship

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Re: Applied Scientist, Alexa Connections - Amazon (Toronto, ON)

Dear Hiring Manager,

I am applying for the Applied Scientist, Alexa Connections role at Amazon. I am a machine learning researcher with a PhD from the University of Trento (2025) and 10+ years across industry and academia, working on speech, audio and sequence models. My most recent published system made real-time decisions in 14 ms on a Raspberry Pi 4 at F1 0.76, 74× faster than the DTW baseline it replaced (JAES 2026).

Here is how my work maps to the role:

- **Speech and ML foundations.** I have built audio DSP and ML pipelines end to end and benchmarked speech models (Wav2Vec2 XLSR-FR vs. DTW-MFCC and 4 other methods over 100,000 scored trials, ROC-AUC 0.822 vs. 0.599).
- **Moving models into production.** On the EU-funded MUSMET project I owned the pipeline from acquisition to deployment with <30 ms inference at >90% F1, and at MAS Holdings I deployed ML systems for manufacturing at scale.
- **Evaluation frameworks.** I design baselines, ablations and error analysis to show what actually improves a model, and I work with LLM and RAG tooling day to day, including Claude Code.

To be direct about the gaps: I have not deployed NER, entity-linking or intent-detection models in production, I have not shipped a conversational or messaging product, and my LLM fine-tuning is tool-level rather than at scale. My NLP work to date is lighter than my audio and time-series work, so I would expect to learn the language-understanding stack from the team.

I am a Canadian permanent resident, eligible to work in Canada without sponsorship, based in Toronto, and available immediately. I would welcome the opportunity to discuss how I can contribute to the Alexa Smart Home science team at Amazon.

Sincerely,

Nishal Stanislaus Silva, PhD